QUALIFICATION FILE - Standard

FULL-STACK DEVELOPER

Type of Training: Long Term Training (LTT)

Type of Qualification: General

Type of Categories: Upskilling

NCrF/NSQF Level: 4.5

Submitted By:
MSME TECHNOLOGY CENTRE
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Section 1: Basic Details

1.	**Qualification Name**	FULL-STACK DEVELOPER	
2.	**Sector(s)**	IT-ITeS	
3.	**Sub-sector(s)**	N/A	
4.	**NCO/ISCO Code(s)** *(If code is not available then mention the same.)*	NCO Code : 2513.01 , ISCO Code : N/A	
5.	**Occupation(s) Name** *(As per Occupational Map or NCO if available.)*	Software Developer	
6.	**Type of Qualification:** New **Qualification has:** Not applicable *(Select none, both or either, as applicable)*	**7. If *Type of Qualification* is Revised, provide Existing Qualification Details** *(As per NQR)* **:** **a. Qualification Name :** N/A **b. NQR Code:** N/A **c. Version:** N/A	
8.	**Qualification, if Developed in Collaboration with MNC*** *Multinational Companies (MNCs) and Leading Indian Enterprises including Original Equipment Manufacturer (OEM), Original Design Manufacturer (ODM) and Value-Added Reseller (VAR) â€' Refer relevant NCVET Guidelines.*	**a. MNC Name:** N/A **b. Qualification Name as per MNC:** N/A	
9.	**National Qualification Register (NQR) Code and Version** *(Issued after NSQC approval)*	**a. NQR Code:** N/A **b. Version:** N/A	**10. NCrF/NSQF Level:** 4.5
11.	**Award (Certificate/Diploma/Advance Diploma/ Any Other)**	Certificate	
12.	**Brief Description of the Qualification**	This qualifications helps learner in the following: Qualified learners should be able to create backend and front end programs to develop the website. Learners are capable to support in designing, installing and maintaining the software systems. Learners are capable in writing the programs for different application Learners are competent in getting employment in IT industries or also become an entrepreneur	

13. Eligibility Criteria for Entry for Students/Trainee/Learner/Employee :

a. Entry Qualification and Relevant Experience:

S.No.	NSQF Level	Academic /Tech Education Completed	Specialization (Academic /Tech Education Completed)	LTT/STT Completed	Specialization (LTT/STT Completed)	Academic /Tech Education Pursuing	Specialization (Academic /Tech Education Pursuing)	Previous NSQF Levels Achieved	Relevant Experience in Years	Relevant Specialization	Remarks
1	4.5	Completed 1st year of UG	N/A	N/A	N/A	N/A	N/A	N/A	No Experience required	N/A	N/A
2	4.5	Completed 3-year diploma after 10	N/A	N/A	N/A	N/A	N/A	N/A	No Experience required	N/A	N/A
3	4.5	Completed 2nd year of 2-year diploma after 12th	N/A	N/A	N/A	N/A	N/A	N/A	No Experience required	N/A	N/A
4	4.5	10th Grade	N/A	1 year NTC plus CITS	N/A	N/A	N/A	N/A	1 year Experience required	N/A	N/A

b. Age *(Please specify age only in case of any legal restrictions)* : N/A

14.	**Training Duration by Modes of Training Delivery** *(Specify training Duration as per selected training delivery modes and as per requirement of the qualification. Refer Blended Learning guidelines and provide details in the Annexure.)*	Training Mode : Blended **a. Minimum Notional Hours** : 600 **b. Maximum Notional Hours** : 600
15.	**Credits Assigned to this Qualification, Subject to Assessment** *(As per National Credit Framework (NCrF). If qualification has Electives or Options, specify both Minimum and Maximum Credits.)*	**a. Minimum Credits** : 20.0 **b. Maximum Credits** : 20.0
16.	**Credits Validity Period** *(As per NCrF SOP)*	Lifelong
	Common Cost Norm Category (I/II/III)	

<table>
<tr><td>17.</td><td>(Wherever applicable, as per CCN notification):</td><td>II</td></tr>
<tr><td>18.</td><td>Any Licensing requirements for Undertaking Training on this Qualification
(Wherever applicable)</td><td>N/A</td></tr>
<tr><td>19.</td><td>Progression Path after Attaining the Qualification
(Please show Professional and Academic progression, as applicable. In case of multiple progression options, mention them separately, including cross sectoral progression.)</td><td>Horizontal Path : Web Developer
Vertical Path : PHP Developer
Academic progression : N/A
NSQF Level: 4.5</td></tr>
<tr><td>20.</td><td>Content Availability Timeline for this qualification
(Within 3 months from NSQC approval date)</td><td>a. Current qualification:
<table>
<tr><th>S.No.</th><th>Language in which MC is being submitted</th><th>Student's /Learner's Handbook (Content Availability Timeline)</th><th>Teacher's /Trainer's Guide (Content Availability Timeline)</th><th>Digital Content (Content Availability Timeline)</th><th>Any Other (Content Availability Timeline)</th><th>If any other (Content Availability Timeline)</th></tr>
<tr><td>1</td><td>English</td><td>N/A</td><td>N/A</td><td>N/A</td><td>N/A</td><td>N/A</td></tr>
</table></td></tr>
<tr><td>21.</td><td>Is similar Qualification(s) available on NQR-if yes, provide details of similar qualifications with justification for developing this qualification</td><td>No
If yes, NQR Code: N/A
Version: N/A
Justification: N/A</td></tr>
<tr><td>22.</td><td>Is the Job Role Amenable to Persons with Disability</td><td>No
If Yes, specify applicable type of Disabilities: N/A</td></tr>
<tr><td>23.</td><td>Is this Qualification for PwD ?</td><td>No</td></tr>
<tr><td>24.</td><td>List Action Plan/Strategy for Encouraging Active Participation of Women</td><td>Seats are reserve as per Govt. norms.</td></tr>
<tr><td>25.</td><td>Are Greening/ Environment Sustainability Aspects Covered
(If yes, specify the Component which covers it, else provide rationale for not including this aspect in the qualification.)</td><td>Yes
Component Details: MSME/ES/01
Rationale: N/A</td></tr>
</table>

26.	**Is Qualification Suitable to be Offered in Schools/Colleges**	Colleges , Schools	
27.	**Name and Contact Details of Submitting / Awarding Body SPOC** *(In case of Cross-Sectoral or Multi-Skill qualification provide details of both Lead AB and Supporting ABs.)*	**Name:** PRANAB KUMAR CHOWDHURY **Email:** pkccttc@gmail.com **Contact No.:** 8763814900 **Website:** https://www.dcmsme.gov.in	
28.	**Date of Approval by NSQC:** 30/04/2024	**29. Validity Duration:** N/A	**30. Next Review Date:** N/A

Section 2: Module Summary

NOS(s)/MC(s)/NC(s)/Module of Qualifications

NOS- *National Occupational Standards* ***MC***- *Micro-Credential* ***NC***- *Nano-Credential*

Specify the training duration and assessment criteria at NOS/MC/NC/Module level. For further details refer to the curriculum document.

Formative assessment marks of the qualification to be included in result calculations: Yes

(Formative assessment for a qualification is optional. If the check box is checked, then the specified Formative assessment marks are to be included in the ***Total****.)*

Note: For all hands-on, practical, and viva assessments, video evidence needs to be captured and maintained through the assessment platform.

Compulsory Components:

Th. - Theory **Pr.** - Practical **OJT Man.** - On the Job Mandatory Training **Proj.** - Project

Mandatory:

S.No.	Component Details			LU Group	Core/ Non-Core	NCrF /NSQF Level	Training Duration (Hours)				Assessment Marks (As applicable)							
	Type (NOS/ MC /NC/ Module)	Name	Code				Theory	Practical	Total (Excluding OJT)	OJT	Formative Assessment	Summative Assessment				Total	Min. Pass (%)	Weightage (%)
												Theory	Practical	Project	Viva			
1	NOS	Deploying and Hosting Application with Docker and Kubernetes	N/A	-	Core	4.5	10.0	80.0	90.0	0.0	0.0	100.0	100.0	0.0	0.0	200.0	-	50
2	NOS	Mastering UI design with Angular	N/A	-	Core	4.5	10.0	50.0	60.0	0.0	0.0	100.0	100.0	0.0	0.0	200.0	-	50
		Working Back-End design																

3	NOS	with Spring Boot and Hibernate Framework	N/A	-	Core	4.5	20.0	190.0	210.0	0.0	0.0	100.0	100.0	0.0	0.0	200.0	-	50
4	NOS	Gaining knowledge in Oracle and MySQL Database Management System	N/A	-	Core	4.5	10.0	80.0	90.0	0.0	0.0	100.0	100.0	0.0	0.0	200.0	-	50
5	NOS	Understanding the Concepts of C and Python Programming	N/A	-	Core	4.5	10.0	50.0	60.0	0.0	0.0	100.0	100.0	0.0	0.0	200.0	-	50
Total							60.0	450.0	510.0	0.0	0.0	500.0	500.0	0.0	0.0	1000.0		

Employability skills:

S.No.	Component Details			LU Group	Core/ Non-Core	NCrF /NSQF Level	Training Duration (Hours)				Assessment Marks (As applicable)							
	Type (NOS/ MC /NC/ Module)	Name	Code				Theory	Practical	Total (Excluding OJT)	OJT	Formative Assessment	Summative Assessment				Total	Min. Pass (%)	Weightage (%)
												Theory	Practical	Project	Viva			
1	NOS	Employability Skill	N/A	-	Non Core	4.5	30.0	0.0	30.0	0.0	0.0	100.0	0.0	0.0	0.0	100.0	-	40
Total							30.0	0.0	30.0	0.0	0.0	100.0	0.0	0.0	0.0	100.0		

Elective:

Elective Table : N/A

Option:

Option Table : N/A

Additional OJT (Hours) (Yes /No)	Yes
Additional OJT (Hours)	60.0

Assessment - Minimum Qualifying Percentage

Minimum Pass Percentage: Aggregate at Qualification / NOS level - Theory and Practical

Aggregate at Qualification Level: N/A %

NOS/MC/NC/Module-wise: N/A

Theory and Practical Aggregate at Qualification Level:
Theory : 40 % and Practical : 60 %

Section 3: Skill Taxonomy

Master Data For The Sector Selected In Basic Details Is Not Available.

Section 4: Training Related

1. **Trainer's Qualification and experience in relevant sector** (in years)
(As per NCVET guidelines. If multiple combinations of qualification and experience are envisaged, mention each combination)

N/A

Trainer's Certification:

S.No.	Domain Certification	Platform Certification
1	YES	YES

2. **Master Trainer's Qualification and experience in relevant sector** (in years)
(As per NCVET guidelines. If multiple combinations of qualification and experience are envisaged, mention each combination)

S.No.	Minimum Educational Criteria	Specialization	Relevant Industry Experience Years	Relevant Industry Experience Specialization	Relevant Training Experience Years	Relevant Training Experience Specialization	Remarks
1	Diploma	Degree in Computer Science	0.0	0	2.0	0	Diploma / Degree in Computer Science/ IT /Any Graduate / or Equivalent with Practical skills and knowledge required in the relevant job role at least one level higher i. e. Level 5 and above in related field and minimum 2 years of

									experience in Tool Room/ Technology Centre of MSME or any reputed industry will become a trainer, or in accordance with the TOT guideline of NCVET.
		Master Trainer's Certification:							
3.	**Tools and Equipment Required for Training**	Yes							

Section 5: Assessment Related

1. **Assessor's Qualification and experience in relevant sector** (in years)
(As per NCVET guidelines. If multiple combinations of qualification and experience are envisaged, mention each combination)

S.No.	Minimum Educational Criteria	Specialization	Relevant Industry Years	Relevant Industry Specialization	Relevant Training Years	Relevant Training Specialization	Remarks
1	Diploma	Degree in Computer Science	0.0	0	3.0	0	Diploma / Degree in Computer Science/ IT /Any Graduate / or Equivalent with 3 years of experience in IT/CS/ Training/ IT from Tool Room/ Technology Centre of MSME or any reputed industry and Only (TOA) certified assessors will be able to conduct the assessments.

Assessor's Certification:

S.No.	Domain Certification	Platform Certification
1	YES	YES

S.No.	Minimum Educational	Specialization	Relevant Industry	Relevant Industry	Relevant Training	Relevant Training	Remarks

			Criteria		Years	Specialization	Years	Specialization	
2.	**Proctor's Qualification and experience in relevant sector** (in years) *(As per NCVET guidelines. If multiple combinations of qualification and experience are envisaged, mention each combination)*	1	Degree in Computer Science	Degree in Computer Science	0.0	0	5.0	0	Degree in Computer Science/ IT /Any Graduate / or Equivalent with 5 years of experience in IT/ Training/ IT Department from Tool Room/ Technology Centre of MSME or any reputed industry.

		S.No.	Minimum Educational Criteria	Specialization	Relevant Industry Years	Relevant Industry Specialization	Relevant Training Years	Relevant Training Specialization	Remarks
3.	**Master Assessor's/Proctor's Qualification and experience in relevant sector** (in years) *(As per NCVET guidelines. If multiple combinations of qualification and experience are envisaged, mention each combination)*	1	Diploma	Degree in Computer Science	0.0	0	3.0	0	Diploma / Degree in Computer Science/ IT /Any Graduate / or Equivalent with 3 years of experience in IT/CS/ Training/ IT from Tool Room/ Technology Centre of MSME or any reputed industry and Only (TOA) certified assessors will be able to conduct

								the assessments.

Master Assessor's Certification:

S.No.	Domain Certification	Platform Certification
1	YES	YES

4.	**Assessment Mode** *(Specify the assessment mode- refer NCVET Guidelines)*	Assessment Mode : Blended
5.	**Tools and Equipment Required for Assessment**	Same as for training

Section 6: Evidence of the need for the Qualification

1. **Latest Skill Gap Study by AB/Indian/Global Organisations** (Study not older than 3 years) **(Yes/No):** Yes

2. **Latest Indian/Global Market, Research Reports, or any other source** (Not older than 2 years) **(Yes/No):** Yes

3. **Government /Industry initiatives/ requirement** (Yes/No) **:** No

4. **Number of Industry validation provided:**

Jan Samridhi Dumka

5. **Estimated number of learners to be trained:**

S.No.	Year	Total Estimated Traning Number	Total Estimated Employment Opportunities	Women Estimated Traning Number	Women Estimated Employment Opportunities	Persons with Disabilities (PwDs) Estimated Traning Number	Persons with Disabilities (PwDs) Estimated Employment Opportunities
1	2024-2025	2000	1600	400	420	0	0
2	2025-2026	2500	2000	500	400	0	0
3	2026-2027	3000	2400	600	480	0	0
Total		7500	6000	1500	1300	0	0

6. **Evidence of Concurrence/Consultation with Line Ministry/State Departments :** Yes

Annexure: Evidence of Level

NSQF Domain	Key requirements of the job role/ outcome of the qualification	How the job role/ outcomes relate to the NSQF Level Descriptor
Professional Theoretical Knowledge/Process	Possesses knowledge in multidisciplinary contexts, broadly, within the chosen fields of technology/ skills/ job role. Deeper knowledge and understanding of specialized field of technology / skills/ job role and its underlying principles, Acquired specialized knowledge and a range of cognitive and practical skills to accomplish tasks like basic design, prototyping, testing so as to solve a problem by selecting appropriate information, methods, tools, and materials.Multidisciplinary and specialized knowledgePossesses broad and deep knowledge and skills to solve problems in specialized fields.	Analyze users needs and then design, test, and developsoftware to meet those needs. Recommend softwareupgrades for customers existing programs and systems.Create a variety of models and diagrams (such asflowcharts) that the learner understand how to writesoftware code. Ensure that a program continues tofunction normally through software maintenance andtesting
Professional and Technical Skills/ Expertise/ Professional Knowledge	Demonstrates cognitive specialised professional and technical skills required for performing and accomplishing difficult tasks relating to the chosen field /s of technology / skills/ job role; successfully applying techniques in routine or non-routine, Possesses a range of professional and technical skills, displays clarity of knowledge and practice in broad range of activities/ tasks. Project Management Skills, Skill to clearly identify the relevant tools or sometimes improvise the available tools and techniques; and has advance knowledge of materials in difficult situations and different contexts. Very good in data collecting organizing information, analysis and communication of results for informed decision making.Range of skills along with specialized domain skillsExcellent cognitive skills and technical prowess utilized to perform complex tasks with ease, project management expertise, and adept at data analysis for informed decisionmaking.	The Developers must analyze users needs and then designsoftware to meet those needs.Developers must understand computer capabilities andprogramming languages in order to design effectivesoftware.Creativity Developers are the creative minds behind newcomputer software. Interpersonal skills. Software developers must be able towork well with others who contribute to designing,developing, and programming successful software.The developers are in charge of software from beginningto end, they must be able to solve problems that arisethroughout the design process.
Employment Readiness & Entrepreneurship Skills & Mind-set /Professional Skill	Possesses excellent oral and written communication and collaboration skills for clearly taking the vision of the leaders to the shop floor level workforce. Possesses very good Digital, Financial and Legal Literacy to use them effectively, Has a good understanding the constitutional, humanistic, ethical, and moral values. Organisation and Time Management, Very good in complex calculations, and mathematical and financial analysis skills for applied solutions. Has well informed practical understanding of the social, political and work environment. Exercise self management within the work contexts. Emotional Intelligence, May have an entrepreneurial Mind set for creating a startup/ small businesses and its end to end managementEntrepreneurial mindset, self managementA versatile professional with excellent communication, digital and financial literacy, ethical values, self management and may have entrepreneurial mindset	Computer applications software engineers analyze userneeds and design, construct, and maintain generalcomputer applications software or specialized utilityprograms. These engineers use different programminglanguages, depending on the purpose of the program. Theprogramming languages most often used are C, C++,python, and Java. Some software engineers develop bothpackaged systems and systems software, or createcustomized applications as per need of IT Industry
Broad Learning Outcomes/Core Skill	Demonstrates a wide range of specialized professional and technical skill in broad range of activity involving standard and non-standard practices. Apply the acquired specialized knowledge and a range of cognitive and practical skills to accomplish tasks like basic design, prototyping, testing so as to solve problems by selecting appropriate information, methods, tools, and materials. Communication and collaboration skills to act as a layer between the senior management and workforce/ shopfloor. Should be able to listen and understand properly and present complex information in a clear and concise manner, make judgement and take decision, based on the analysis and evaluation of information, for determining solutions to a variety of unpredictable problems associated with the chosen fields of learning, Takes responsibility for the nature and quality of outputs. Is able to work on processes to improve the quality of outputs, Can analyze and synthesize ideas, Uses discretion and judgement over a range of known and innovative responses to familiar and unfamiliar problems and issues.Judgement /	As a Software engineer he/she have ability to determinewhat code is necessary to achieve some specifiedfunctionality, particularly the higher-level structure ororganization of the code. Software Engineers ability toanalyze the behavior of code to diagnose a problem andfind the underlying cause. This includes but is not limitedto using a debugger

	decision making specializedA skilled professional with technical expertise, adept at solving complex problems and improving output.	
Responsibility	Is accountable for determining and achieving personal and /or group tangible outcomes, Handles/ / adapts/ accommodates change requirements and change management at the ground/ shop floor level. Team Building, Manages processes and procedures within broad parameters for defined activities. Supervises the routine work of others, takes the required responsibility for the evaluation and improvement of work or study activities. At level 4.5 the candidate is Highly skilled and works as a Junior Technical Supervisor. Constantly motivates, guides, mentors and trainsthe workforce. At level 5.0 the candidate is a Technical supervisor or junior/ deputy manager. Is responsible for managing an independent work unit/ shop floor/ section/ business activity/ assignmentTeam leader Junior technical supervisor, Technical supervisor or junior/ deputy manager.Highly skilled Technical Supervisor responsible for achieving tangible outcomes, managing change, building teams, and mentoring the workforce	Programming computer controls, Working with businessanalyst to develop project implementation and developplans including user interfaces, Modifying IT systemsalready in use, Writing documentation, Developing Codingand debugging across a variety of products, Testing newSoftwares for compatibility with other existing softwares,Fixing any technical problems while testing, Designing,prototyping and implementing graphical user interfaces,etc. Thus the curriculum of the qualification is so designedto take the responsibility for own work and learning as wellas other team members work and learning.

Annexure: Tools and Equipment (Lab Set-Up)

List of Tools and Equipment

Batch Size : 20

S.No.	Tool or Equipment Name	Specification	Tool or Equipment ratio per the number of learners	Remarks
1	Computer System	Industry Standard	1:1	-
2	Programing software Turbo C , C++ / DEV C++	Industry Standard	1:1	-
3	Programing software JAVA JDK, Eclipse	Industry Standard	1:1	-
4	Programing software Apache Tomcat, MySQL workbench , Microsoft Office Front Page , VS code , Oracle XE univ, STS, POSTMAN,GIT	Industry Standard	1:1	-

Classroom Aids

The aids required to conduct sessions in the classroom are:

S.No.	Aids required to conduct sessions in the classroom / Equipment Name
1	NA

Annexure: Industry Validations Summary

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Annexure: Training and Employment Details

Training and Employment Projections:

S.No.	Year	Total Estimated Traning Number	Total Estimated Employment Opportunities	Women Estimated Traning Number	Women Estimated Employment Opportunities	Persons with Disabilities (PwDs) Estimated Traning Number	Persons with Disabilities (PwDs) Estimated Employment Opportunities
1	2024-2025	2000	1600	400	420	0	0
2	2025-2026	2500	2000	500	400	0	0
3	2026-2027	3000	2400	600	480	0	0
Total		7500	6000	1500	1300	0	0

Training, Assessment, Certification, and Placement Data for previous versions of qualifications:

										Total Persons	Total Persons	Total Persons	Total Persons

S.No.	Year	Total Candidates Trained	Total Candidates Assessed	Total Candidates Certified	Total Candidates Placed	Total Women Trained	Total Women Assessed	Total Women Certified	Total Women Placed	with Disabilities (PwDs) Trained	with Disabilities (PwDs) Assessed	with Disabilities (PwDs) Certified	with Disabilities (PwDs) Placed
1	2021-2022	0	0	0	0	0	0	0	0	0	0	0	0
2	2022-2023	0	0	0	0	0	0	0	0	0	0	0	0
3	2023-2024	0	0	0	0	0	0	0	0	0	0	0	0

Annexure: Blended Learning

Blended Learning Estimated Ratio and Recommended Tools:

Refer NCVET Guidelines for Blended Learning for Vocational Education, Training and Skilling available on:
https://ncvet.gov.in/wp-content/uploads/2023/01/Guidelines-for-Blended-Learning-for-Vocational-Education-Training-Skilling.pdf

S.No.	Components of the MC	List Recommended Tools for all Selected Components	Offline:Online Ratio
1	Theory/ Lectures - Imparting theoretical and conceptual knowledge	Books/ e-books, Presentations, Reference Material , Audio / VideoModules with installation step, settings and configuration SelfLearning Videos /Broadcasts /Mobile Learning /Curated Digitalcontent	40:60
2	Imparting Soft Skills, Life Skills, and Employability Skills /Mentorship to Learners	Self-Learning Videos , Broadcasts, Mobile Learning , Curated Digitalcontent	40:60
3	Showing Practical Demonstrations to the learners	Installation and path set of software, website design, databaseconnectivity, GUI application, Video Content, E-Resource library.	100:0
4	Imparting Practical Hands-on Skills/ Lab Work /workshop/ shop floor training	Turbo C++, DEV C++, My Eclipse , Apache Tomcat, MySQLworkbench , Microsoft Office Front Page , VS code , Oracle XE univ	100:0
5	Tutorials/ Assignments/ Drill/ Practise	Online Question Bank, Mobile Quick test app, MCQ based tests,Practical Test on Machines	40:60
6	Proctored Monitoring/ Assessment/ Evaluation/ Examinations	Assessment engine for Essays, Up-loadable file examinations,Mock test sessions	40:60

7	On the Job Training (OJT)/ Project Work Internship /Apprenticeship Training	NA	100:0

Annexure: Detailed Assessment Criteria

Ensure the Module Summary Assessment Marks are mapped to the Detailed Assessment Criteria.
Detailed assessment criteria for each NOS/MC/NC/Module are as follows (Refer NCVET Guidelines):

Component Type(NOS /MC/NC /Module)	NOS/MC/NC /Module Name	Assessment Criteria for Performance Criteria /Learning Outcomes	Formative Assessment	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS	Employability Skill	Explain the major applications of MS Office	-	-	-	-	-
NOS	Employability Skill	Explain the different types of e-commerce	-	-	-	-	-
NOS	Employability Skill	List the benefits of e-commerce for retailers and customers	-	-	-	-	-
		Discuss how the Digital					

NOS	Employability Skill	India campaign will help boost e-commerce in India	-	-	-	-	-
NOS	Employability Skill	Write applications pertaining to various matters	-	-	-	-	-
NOS	Employability Skill	Explain power of positive attitude and Importance of commitment	-	-	-	-	-
NOS	Employability Skill	Explain motivation and the Ways to motivate oneself and Personal goal setting	-	-	-	-	-
NOS	Employability Skill	Explain the Effective and Level of Communication	-	-	-	-	-
NOS	Employability Skill	Explain communication and Significance of technical communication?	-	-	-	-	-

NOS	Employability Skill	Explain the methods of listening Skills.	-	-	-	-	-
NOS	Employability Skill	Explain the differences between bio-data, CV and Resume.	-	-	-	-	-
NOS	Employability Skill	Explain verbal and non-verbal Communication	-	-	-	-	-
NOS	Employability Skill	Explain how to face an interview.	-	-	-	-	-
NOS	Employability Skill	Explain team work, group work, team formation process	-	-	-	-	-
NOS	Employability Skill	How to Minimize the team conflicts	-	-	-	-	-
NOS	Employability Skill	Explain Ethics and values	-	-	-	-	-
NOS	Employability Skill	Explain the concept of entrepreneurship, and entrepreneursh	-	-	-	-	-

		ip v/s Management					
NOS	Employability Skill	Explain the process of project report preparation for setting up a new business	-	-	-	-	-
NOS	Employability Skill	Explain the role of various schemes and institute for selfemployment i.e. MSME, DIC, NSIC, SIDBI etc.	-	-	-	-	-
NOS	Employability Skill	Role of financial institution to support startup	-	-	-	-	-
NOS	Employability Skill	Discuss the importance of saving money	-	-	-	-	-
NOS	Employability Skill	Discuss the main types of bank accounts	-	-	-	-	-
NOS	Employability Skill	Differentiate between fixed and variable costs	-	-	-	-	-
		Describe the					

NOS	Employability Skill	different types of insurance products	-	-	-	-	-
NOS	Employability Skill	Discuss the main types of electronic funds transfers	-	-	-	-	-
Total			**0.0**	**100.0**	**0.0**	**0.0**	**0.0**
NOS	Deploying and Hosting Application with Docker and Kubernetes	Explain the concepts of cloud computing.	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Understand the importance of virtualization in distributed computing and how this has enabled the development of Cloud Computing.	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Analyze the performance of Cloud Computing.	-	-	-	-	-

NOS	Deploying and Hosting Application with Docker and Kubernetes	Understand the concept of Cloud Security	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Learn the Concept of Cloud Infrastructure Model.	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Understand about the concept of AWS.	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Able to create AWS Account	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Demonstrate the concept of IAM and AWS CLI.	-	-	-	-	-
	Deploying and Hosting Application	Able to create					

NOS	with Docker and Kubernetes	an EC2 Instance	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Understand the concept of Amazon RDS	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Able to deploy Spring Boot application on AWS using Elastic Beanstalk.	-	-	-	-	-
NOS	Deploying and Hosting Application with Docker and Kubernetes	Able to explain the concepts of Docker and Kubernetes.	-	-	-	-	-
Total			**0.0**	**100.0**	**100.0**	**0.0**	**0.0**
NOS	Mastering UI design with Angular	Demonstrate the Structure of an html Page.	-	-	-	-	-
NOS	Mastering UI design with Angular	Demonstrate different types of basic html tag.	-	-	-	-	-

NOS	Mastering UI design with Angular	Demonstrate the use of list and table tag.	-	-	-	-	-
NOS	Mastering UI design with Angular	Demonstrate the different types of form tag and their uses for User interfaces.	-	-	-	-	-
NOS	Mastering UI design with Angular	Demonstrate External Style Sheets, Internal Style Sheets, and Inline Style, The class selector, div and span tag.	-	-	-	-	-
NOS	Mastering UI design with Angular	Run a HTML Program using manual process and Adobe Dream weaver IDE.	-	-	-	-	-
NOS	Mastering UI design with Angular	Design and develop a web page using heading, font, image, marquee tag.	-	-	-	-	-
	Mastering UI	Develop a web page using order					

NOS	design with Angular	list, unordered list and definition list.	-	-	-	-	-
NOS	Mastering UI design with Angular	Design a Student registration form that will take all student Data including their photo and CV.	-	-	-	-	-
NOS	Mastering UI design with Angular	Design a web page using html and CSS with batter look and fill and then understand the concept of class and id in CSS	-	-	-	-	-
NOS	Mastering UI design with Angular	Design web page using Angular JS	-	-	-	-	-
Total			**0.0**	**100.0**	**100.0**	**0.0**	**0.0**
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Explain the concept of control statement and looping	-	-	-	-	-
		Explain the					

NOS	Working Back-End design with Spring Boot and Hibernate Framework	concept of class and object and using object access the member of a class.	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	xplain Different ways of representing a method passing primitive data types to method, method overloading, Types of method: instance, static, recursive, factory method. Passing object to method, passing array to method	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate	Explain the use of inheritance in programming	-	-	-	-	-

	Framework	Language.					
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Create a Thread by using Thread class and Runnable interface, Lifecycle of Thread, Thread class constructor and methods, Thread priorities, thread group, multithreading and synchronization	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Explain the Concept of file	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	xplain the concept of TCP/IP protocol, sockets, Knowing IP Address, URL and URL Connection class	-	-	-	-	-

NOS	Working Back-End design with Spring Boot and Hibernate Framework	Develop GUI application using swing through Net Beans IDE.	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Explain the Architecture of JDBC and Types of JDBC Driver.	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Explain the Concept of Statement, Prepared Statement	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Execute stored procedure and function using Callable Statement Interface.	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Explain the concept of Spring and Spring Boot.	-	-	-	-	-
	Working	Develop					

NOS	Back-End design with Spring Boot and Hibernate Framework	CRUD application using Hibernate with Data JPA.	-	-	-	-	-
NOS	Working Back-End design with Spring Boot and Hibernate Framework	Explain the concept of Restful web services.	-	-	-	-	-
Total			**0.0**	**100.0**	**100.0**	**0.0**	**0.0**
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Explain the Difference between DBMS and File Base System, DBMS Architecture.	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Explain the concept of Data Definition Language (DDL)	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database	Explain the concept of Data Manipulation Language (DML) to	-	-	-	-	-

	Management System	manipulate over Data.					
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Explain the Concept of Different types of constraints	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Demonstrate how two or more table can be join using different types of join operation.	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Explain the concept of stored procedure and function.	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Install Data Base softwares like MYSQL, SQL Server, Oracle and DB2.	-	-	-	-	-
		Students will be able to execute SQL					

NOS	Gaining knowledge in Oracle and MySQL Database Management System	queries using Data Definition Language DDL commands such as CREATE ALTER and DROP to create modify and delete database tables.	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Students will be able to use SQL commands to view add change and delete data in a database.	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Students will be able to modify tables by adding different constraints.	-	-	-	-	-
NOS	Gaining knowledge in Oracle and MySQL Database	Explain a table and then join those table by using different types	-	-	-	-	-

	Management System	of joining operation.					
NOS	Gaining knowledge in Oracle and MySQL Database Management System	Able to create a store procedure by using both IN and OUT parameter and then execute them	-	-	-	-	-
Total			**0.0**	**100.0**	**100.0**	**0.0**	**0.0**
NOS	Understanding the Concepts of C and Python Programming	Demonstrate the structure of C program	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Demonstrate different data types, variable and keywords and identifier, operator and its uses.	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Explain different control statement, types of loop and go to, continue keyword	-	-	-	-	-
	Understanding	Explain the					

NOS	the Concepts of C and Python Programming	way to declare, define and call a function	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Explain pointers-concepts, initialize pointer variables, Character pointers and functions, pointers to pointers	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Explain types of array, store, sort and search data within it.	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Explain string and its pre-defined function.	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Explain structure and union	-	-	-	-	-
NOS	Understanding the Concepts of C and	Explain file handling for Permanente	-	-	-	-	-

	Python Programming	data store process					
NOS	Understanding the Concepts of C and Python Programming	Demonstrate the Python program	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Demonstrate the use of print statements	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Demonstrate the knowledge of operator, control system and looping	-	-	-	-	-
NOS	Understanding the Concepts of C and Python Programming	Demonstrate the knowledge of functions in python	-	-	-	-	-
Total			**0.0**	**100.0**	**100.0**	**0.0**	**0.0**
Grand Total			**0.0**	**600.0**	**500.0**	**0.0**	**0.0**